

60.5.4 Trace funnel (CSTF)

The trace funnel is a CoreSight component that combines the ATB buses from two trace sources into one single ATB. The CSTF has two ATB slave ports, and one ATB master port. An arbiter selects the slave ports according to a programmable priority.

The slave ports are connected as follows:

- S0: Cortex-M7 ETM
- S1: Cortex-M7 ITM

The CSTF registers allow the slave ports to be individually enabled, and their priority settings to be configured. The priorities can be modified only when trace is disabled. The arbitration works as follows:

- The arbiter selects the slave port with the highest assigned priority that has data valid
- Up to *min_hold_time* transfers are passed from the selected slave to the master port, where *min_hold_time* is programmable in the CONTROL register.
- A new arbitration is then performed

High priority should be assigned to slave ports connected to sources with a small amount of buffering, or where data loss can not be tolerated. Low priority should be assigned to less critical sources or those with large buffers.

For more information on the ATB Funnel CoreSight component, refer to the Arm® CoreSight™ SoC-400 Technical Reference Manual [2].

Trace funnel registers

CSTF control register (CSTF_CTRL)

Address offset: 0x000

Reset value: 0x0000 0300

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	MIN_HOLD_TIME[3:0]				Res.	Res.	Res.	Res.	Res.	Res.	ENS1	ENS0
				rw	rw	rw	rw							rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:8 **MIN_HOLD_TIME[3:0]**: Number of transactions between arbitrations.

0x0: 1 transaction

:

0xE: 15 transactions

0xF: Reserved

Bits 7:2 Reserved, must be kept at reset value.

Bit 1 **ENS1**: S1 slave port enable

0: Disable port

1: Enable port

Bit 0 **ENS0**: S0 slave port enable

0: Disable port

1: Enable port

CSTF priority register (CSTF_PRIORITY)

Address offset: 0x004

Reset value: 0x0000 0688

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PRIPORT1[2:0]			PRIPORT0[2:0]		
										rW	rW	rW	rW	rW	rW

Bits 31:6 Reserved, must be kept at reset value.

Bits 5:3 **PRIPORT1[2:0]**: S1 slave port priority

0: Highest priority

:

7: Lowest priority

Bits 2:0 **PRIPORT0[2:0]**: S0 slave port priority

0: Highest priority

:

7: Lowest priority

CSTF claim tag set register (CSTF_CLAIMSET)

Address offset: 0xFA0

Reset value: 0x0000 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMSET[3:0]			
												rW	rW	rW	rW

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **CLAIMSET[3:0]**: Set claim tag bits

Write:

0000: No effect

xxx1: Set bit 0

xx1x: Set bit 1

x1xx: Set bit 2

1xxx: Set bit 3

Read:

0xF: Indicates there are four bits in claim tag

CSTF claim tag clear register (CSTF_CLAIMCLR)

Address offset: 0xFA4

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMCLR[3:0]			
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **CLAIMCLR[3:0]**: Reset claim tag bits

Write:

0000: No effect

xxx1: Clear bit 0

xx1x: Clear bit 1

x1xx: Clear bit 2

1xxx: Clear bit 3

Read: Returns current value of claim tag

CSTF lock access register (CSTF_LAR)

Address offset: 0xFB0

Reset value: N/A

[illegible]

Bits 31:0 **ACCESS_W[31:0]**: CSTF register write access enable

The field enables write access to some CSTF registers by processor cores (debuggers do not need to unlock the component).

0xC5ACCE55: Enable write access

Other values: Disable write access

CSTF lock status register (CSTF_LSR)

Address offset: 0xFB4

Reset value: 0x0000 0003

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LOCKT YPE	LOCKG RANT	LOCKE XIST
													r	r	r

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **LOCKTYPE**: Size of the CSTF_LAR register

0: 32-bit

Bit 1 **LOCKGRANT**: Current status of lock

This bit always returns zero when read by an external debugger.

0: Write access is permitted

1: Write access is blocked. Only read access is permitted.

Bit 0 **LOCKEXIST**: Existence of lock control mechanism

The bit indicates whether a lock control mechanism exists. It always returns zero when read by an external debugger.

0: No lock control mechanism exists

1: Lock control mechanism is implemented

CSTF authentication status register (CSTF_AUTHSTAT)

Address offset: 0xFB8

Reset value: 0x0000 000A

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SNID[1:0]		SID[1:0]		NSNID[1:0]		NSID[1:0]	
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:6 **SNID[1:0]**: Security level for secure non-invasive debug
0x0: Not implemented

Bits 5:4 **SID[1:0]**: Security level for secure invasive debug
0x0: Not implemented

Bits 3:2 **NSNID[1:0]**: Security level for non-secure non-invasive debug
0x2: Disabled
0x3: Enabled

Bits 1:0 **NSID[1:0]**: Security level for non-secure invasive debug
0x2: Disabled
0x3: Enabled

CSTF CoreSight device identity register (CSTF_DEVID)

Address offset: 0xFC8

Reset value: 0x0000 0024

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SCHEME[3:0]				PORTCNT[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **SCHEME[3:0]**: Priority scheme
0x2: Static priority

Bits 3:0 **PORTCNT[3:0]**: Number of input ports connected
0x4: Four input ports

CSTF CoreSight device type identity register (CSTF_DEVTYPE)

Address offset: 0xFCC

Reset value: 0x0000 0012

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DEVTYPEID[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **DEVTYPEID[7:0]**: Device type identifier
0x12: Trace funnel

CSTF CoreSight peripheral identity register 4 (CSTF_PIDR4)

Address offset: 0xFD0

Reset value: 0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **4KCOUNT[3:0]**: Register file size
0x0: Register file occupies a single 4 Kbyte region

Bits 3:0 **JEP106CON[3:0]**: JEP106 continuation code
0x4: Arm® JEDEC code

CSTF CoreSight peripheral identity register 0 (CSTF_PIDR0)

Address offset: 0xFE0

Reset value: 0x0000 0008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PARTNUM[7:0]**: Part number field, bits [7:0]
0x08: CSTF part number

CSTF CoreSight peripheral identity register 1 (CSTF_PIDR1)

Address offset: 0xFE4

Reset value: 0x0000 00B9

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **JEP106ID[3:0]**: JEP106 identity code field, bits [3:0]

0xB: Arm® JEDEC code

Bits 3:0 **PARTNUM[11:8]**: Part number field, bits [11:8]

0x9: CSTF part number

CSTF CoreSight peripheral identity register 2 (CSTF_PIDR2)

Address offset: 0xFE8

Reset value: 0x0000 003B

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]		
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVISION[3:0]**: Component revision number

0x3: r1p1

Bit 3 **JEDEC**: JEDEC assigned value

1: Designer ID specified by JEDEC

Bits 2:0 **JEP106ID[6:4]**: JEP106 identity code field, bits [6:4]

0x3: Arm® JEDEC code

CSTF CoreSight peripheral identity register 3 (CSTF_PIDR3)

Address offset: 0xFEC

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVAND[3:0]**: Metal fix version

0x0: No metal fix

Bits 3:0 **CMOD[3:0]**: Customer modified

0x0: No customer modifications

CSTF CoreSight component identity register 0 (CSTF_CIDR0)

Address offset: 0xFF0

Reset value: 0x0000 000D

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[7:0]**: Component ID field, bits [7:0]

0x0D: Common ID value

CSTF CoreSight component identity register 1 (CSTF_CIDR1)

Address offset: 0xFF4

Reset value: 0x0000 0090

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **CLASS[3:0]**: Component ID field, bits [15:12] - component class
 0x9: CoreSight component

Bits 3:0 **PREAMBLE[11:8]**: Component ID field, bits [11:8]
 0x0: Common ID value

CSTF CoreSight component identity register 2 (CSTF_CIDR2)

Address offset: 0xFF8

Reset value: 0x0000 0005

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[19:12]**: Component ID field, bits [23:16]
 0x05: Common ID value

CSTF CoreSight component identity register 3 (CSTF_CIDR3)

Address offset: 0xFFC

Reset value: 0x0000 00B1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[27:20]**: Component ID field, bits [31:24]
 0xB1: Common ID value

Trace funnel register map and reset values

Table 614. CSTF register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x000	CSTF_CTRL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MIN_HOLD_TIME[3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	ENS1	ENS0
	Reset value																					0	0	1	1							0	0	
0x004	CSTF_PRIORITY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.			PRIORIT1[2:0]		PRIORIT0[2:0]		
	Reset value																											0	0	1	0	0	0	
0x008 to 0xF9C	Reserved	Reserved																																
0xFA0	CSTF_CLAIMSET	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.		CLAIMSET[3:0]			
	Reset value																													1	1	1	1	
0xFA4	CSTF_CLAIMCLR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.		CLAIMCLR[3:0]			
	Reset value																												0	0	0	0		
0xFA8to 0xFAC	Reserved	Reserved																																
0xFB0	CSTF_LAR	ACCESS_W[31:0]																																
	Reset value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
0xFB4	CSTF_LSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																														0	1	1	
0xFB8	CSTF_AUTHSTAT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																													0	0	0	0	
0xFC8	CSTF_DEVID	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																												0	0	1	0	0	
0xFCC	CSTF_DEVTYPE	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																												0	0	0	1	0	0

Table 614. CSTF register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
0xFD0	CSTF_PIDR4	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]	0	0	0	0	0	1	0	0	0					
	Reset value																																							
0xFD4 to 0xFDC	Reserved	Reserved																																						
0xFE0	CSTF_PIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]														
	Reset value																									0	0	0	0	1	0	0	0							
0xFE4	CSTF_PIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]										
	Reset value																									1	0	1	1	1	0	0	1							
0xFE8	CSTF_PIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]									
	Reset value																									0	0	1	1		1	0	1	1						
0xFEC	CSTF_PIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]										
	Reset value																									0	0	0	0	0	0	0	0							
0xFF0	CSTF_CIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]														
	Reset value																									0	0	0	0	1	1	0	1							
0xFF4	CSTF_CIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]										
	Reset value																									1	0	0	1	0	0	0	0							
0xFF8	CSTF_CIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]														
	Reset value																									0	0	0	0	0	1	0	1							
0xFFC	CSTF_CIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]														
	Reset value																									1	0	1	1	0	0	0	1							

60.5.5 Embedded trace FIFO (ETF)

The ETF is a 4 Kbyte memory that captures trace data from two trace sources, namely the ETM and ITM of the CPU core. The ETF is a design configuration of the CoreSight™ trace memory controller component.

The ETF can be used in three modes (selected in the mode register):

1. Hardware FIFO mode

The trace memory is used as a FIFO that is drained through the ATB master interface. Trace data is captured into the trace RAM and when full, the incoming trace stream is stalled. When the Trace buffer is not empty, trace data is drained out through the ATB master interface to the TPIU.

In this mode, the role of the FIFO is to smooth the flow of trace information arriving at the trace port. Since the trace data can be very irregular by nature, the peak data rate can easily exceed the port capability, resulting in an overflow. The ETF allows a steady data rate at the trace port, which can then be sized according to the average rate rather